

Statistic : Chapter 5 : Sampling Distribution

Exercise 5-1

a) mean = 40, std deviation = 3.5, n = 80.

$$\frac{3.5}{\sqrt{80}} = 0.391$$

for ~~z~~ $n = 39.5$

$$z = \frac{39.5 - 40}{0.391} = -1.28$$

For $n = 41.5$

$$z = \frac{41.5 - 40}{0.391} = 3.83$$

$$P(-1.28 < z < 3.83) = P(z < 3.83)$$

$$- P(z < -1.28)$$

$$P = 0.9999 - 0.1003$$

$$= 0.8996$$

b) $z = 1.79$

$$0.391$$

$$P(-1.79 < z < 1.79) = 0.9633 - 0.0367 = 0.9266$$

c) $z = 39 - 40 = -2.56$

$$0.391$$

$$P(z < -2.56) = 0.0052$$

Exercise 5-2

a) mean = 2255, std deviation = 148, n = 90

$$SE = \frac{148}{\sqrt{90}} = 15.61$$

$$z = \pm \frac{25}{15.61} = \pm 1.60$$

$$P(-1.60 < z < 1.60) = 0.9452 - 0.0548 = 0.8904$$

$$= 1 - 0.8904$$

$$= 0.1096$$

b) $z = \frac{20}{15.61} = 1.28$

$$P(z > 1.28) = 1 - P(z < 1.28)$$

$$= 1 - 0.8997$$

$$= 0.1003$$

Exercise 5-3

Mean = 64, std deviation = 0.4, n = 16

$$SE = \frac{0.4}{\sqrt{16}} = 0.1$$

$$\sqrt{16}$$

For $\mu = 63.75$

$$Z = \frac{63.75 - 64}{0.1} = -2.5$$

For $\mu = 64.20$

$$Z = \frac{64.20 - 64}{0.1} = 2.0$$

$$P(Z < -2.5) = 0.0062, P(Z > 2.0) = 0.0228$$

$$= 0.0062 + 0.0228 = 0.0290$$

Exercise 5-4 $p = 0.05, q = 0.95, n = 200$

$$a) SE = \sqrt{\frac{pq}{n}} = \sqrt{\frac{(0.05)(0.95)}{200}} = 0.0154$$

$$Z = \pm \frac{0.03}{0.0154} = \pm 1.95$$

$$P(-1.95 < Z < 1.95) = 0.9744 - 0.0256 = 0.9488$$

b) Greater than $p + 0.05$

$$Z = \frac{0.05}{0.0154} = 3.25$$

$$P(Z > 3.25) = 1 - P(Z < 3.25) = 1 - 0.994 = 0.006$$

c) Less than $p - 0.06$

$$Z = \frac{-0.06}{0.0154} = -3.90$$

$$P(Z < -3.90) = 0.0001$$

d) $p = 0.05, q = 0.95, n = 200$

$$SE = \sqrt{\frac{(0.05)(0.95)}{200}} = 0.0154$$

$$P(Z) = 0.60, Z = 0.25$$

$$Z = \frac{\hat{p} - p}{SE}$$

$$0.25 = \frac{\hat{p} - 0.05}{0.0154}$$

$$\hat{p} = 0.05 + (0.25)(0.0154)$$

$$= 0.05 + 0.00385$$

$$= 0.0539$$

Exercise 5-5

$$p = 0.96, q = 0.04, n = 300$$

$$SE = \sqrt{\frac{(0.96)(0.04)}{300}} = 0.0113$$

$$\begin{array}{lcl} \text{a) } p = 0.04 & \text{b) } z = \frac{\hat{p} - p}{SE} = \frac{0.08 - 0.04}{0.0113} = \frac{0.04}{0.0113} = 3.54 \\ SE = 0.0113 & & \end{array}$$

$$p(z < 3.54) = 0.9998$$

$$\begin{array}{lcl} \text{c) } z = \frac{\hat{p} - p}{SE} = \frac{0.94 - 0.96}{0.0113} = \frac{-0.02}{0.0113} = -1.77 \end{array}$$

$$P(z < -1.77) = 0.0384$$

$$= 1 - 0.0384$$

$$= 0.9616$$